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RESEARCH ARTICLE

Harnessing *S. aureus* tRNA Guanine Transglycosylase for Efficient Site-Specific Covalent Labeling of DNA

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Abstract: Site-specific DNA modification enables the construction of functional nucleic acid architectures for imaging, diagnostics, and information encoding. Bacterial tRNA guanine transglycosylases (TGTs) have been shown to exchange guanine-34 in cognate tRNAs for prequeuosine1 (preQ1) analogs. *Escherichia coli* TGT (EcTGT) has been widely harnessed for site-specific RNA labeling. However, EcTGT exhibits slow labeling kinetics and lower efficiency on DNA. Here, we demonstrate that TGT from *Staphylococcus aureus* (SaTGT) is a superior enzyme for DNA modification. Like EcTGT, SaTGT recognizes only the anticodon stem-loop and efficiently modifies the DNA hairpin dECYMH, whereas EcTGT shows no appreciable modification. Using the *S. aureus* tRNA^{His}-based hairpin as a starting point, we identified the most promising scaffolds. Rational loop optimization yielded dSAH-6, which is labeled by SaTGT with near-quantitative conversion under standard conditions. Side-by-side comparisons revealed that SaTGT outperforms EcTGT in both yield and kinetics across multiple preQ1 derivatives and hairpin sequences. Importantly, the optimized dSAH-6 sequence enables robust labeling at 5′-, 3′-, or internal positions in longer ssDNA constructs and supports efficient multi-site incorporation. These findings significantly expand the DNA-TAG toolbox and highlight how natural variation in TGT substrate promiscuity can be exploited for improved nucleic acid modification technologies.

Introduction

Nucleic acid labeling is essential for creating molecular probes that elucidate biomolecular structure, function, and dynamics both *in vitro* and in living cells[1]. Among the numerous labeling strategies that have been developed, covalent labeling is attractive because it provides a highly stable linkage. Traditional covalent labeling of nucleic acids relies primarily on solid-phase chemical synthesis, which offers great flexibility in generating chemically diverse nucleic acids. However, chemical labeling is often limited to short sequences, requires laborious synthetic and purification processes, and often suffers from low yield[2]. In contrast, chemoenzymatic labeling approaches offer several advantages. Enzymes operate under mild and aqueous conditions, achieve exquisite site specificity through their intrinsic high substrate specificity, and are typically not limited by nucleic acid length[3,4]. Enhancing enzymatic tools for covalent, site-specific nucleic acid labeling under physiological conditions holds tremendous promise for applications in RNA and DNA biology, molecular diagnostics and therapeutics[5,6].

Our group previously developed RNA transglycosylation at guanosine (RNA-TAG) using *Escherichia coli* tRNA guanine transglycosylases (EcTGT) as a tool for the site-specific labeling of RNA. TGTs catalyze the exchange of a guanine at the wobble position of the anticodon loop of queuine-cognate tRNAs with a 7-deazaguanine queuine precursor, preQ$_1$[7]. This enzyme requires only a minimal 17-nucleotide hairpin structure mimicking the tRNA anticodon stem-loop for recognition[8,9] and can label extended RNAs bearing this element[10,11]. Inspired by this mechanism, we have used EcTGT to incorporate preQ$_1$ derivatives functionalized with fluorophores, affinity handles, and bioorthogonal groups into TGT recognition elements. *In vivo*, TGT activity is restricted to RNA substrates. However, recent work by our group has shown that rational engineering of DNA hairpins to minimize steric hindrance in the active site enables EcTGT to recognize and label specific DNA substrates, leading to the development of a method called DNA transglycosylation at deoxyguanosine (DNA-TAG)[12]. Since their inception, RNA- and DNA-TAG have been harnessed for a number of biotechnological applications, including light activated control of translation and CRISPR gene editing[13,14], affinity purification and identification of RNA-protein interactions[15], RNA-protein conjugation[16], mRNA labeling and imaging[17,18], and Northern blotting and single-molecule FISH[12].

DNA-TAG was developed by optimizing DNA hairpin sequences to improve their fit within the TGT active site, reducing steric clashes between the enzyme and the deoxyribose backbone and thymine bases of the DNA substrate[12]. This iterative mutation process, which required testing over 100 variants, is laborious. An alternative approach, in which EcTGT itself was mutagenized to improve its affinity for DNA failed to produce an enzyme that could efficiently incorporate preQ1 derivatives onto DNA hairpins[19]. The strong preference of EcTGT for RNA, and the difficulty in directly engineering EcTGT to accept alternative substrates, inspired us to explore TGTs from other organisms to identify an enzyme with increased promiscuity toward DNA substrates. TGTs are present across all kingdoms of life and exhibit variation in small-molecule substrate specificity. Generally, bacterial TGTs incorporate preQ1 at the wobble position of tRNA anticodon loop. However, evolutionary adaptation has produced notable exceptions. Recent studies have shown that certain bacterial TGTs from intracellular human pathogens, such as *Chlamydia trachomatis*, have shifted their substrate specificity from preQ1 to queuine, while others, such as *Bartonella henselae*, exhibit dual-substrate specificity, accepting both preQ1 and queuine[20,21]. Since queuine is bulkier than preQ1, these shifts imply differences in active-site tolerance that could accommodate varied substrates. Therefore, TGT enzymes from alternative bacterial species may be uniquely suited for the task of DNA labeling, and exploring TGTs from diverse species may facilitate novel nucleic acid modification reactions.

Here we demonstrate the TGT from *Staphylococcus aureus* (SaTGT) can serve as an alternative enzyme for both RNA-TAG and DNA-TAG. Through rational design and optimization of DNA hairpin sequences, we achieved high efficiency incorporation of preQ1 probes into 25-nucleotide DNA hairpins. Next, we characterized the ability of DNA-TAG to label DNA constructs containing one or more recognition sequences. Finally, we compared the labeling kinetics of SaTGT and EcTGT toward an optimized DNA substrate and found that SaTGT exhibited superior performance. SaTGT thus enhances the DNA-TAG toolbox, highlights the differential substrate promiscuity of TGTs from different species, and opens the door to novel DNA and RNA-based applications.

Results and Discussion

We have previously demonstrated that EcTGT can incorporate preQ1 derivatives into both RNA and DNA substrates (Figure 1A, B)[12,22]. In DNA-TAG reactions with EcTGT, the labeling efficiency remains suboptimal with maximal modification typically requiring about 4 hours incubation[12]. The labeling reaction is monitored by a gel-shift assay that exploits the molecular-weight increase upon covalent modification of the substrate with the preQ1 conjugate. Successful enzymatic exchange of guanine for higher molecular weight preQ1-derivatives results in a clear upward shift of the product band relative to the unmodified oligonucleotide (Figure 1C).

Prior studies suggest that TGT may play a role in the virulence and pathogenicity of bacteria[10]. Therefore, we selected TGT from *S. aureus*, a major Gram-positive bacterial pathogen responsible for a broad spectrum of clinical infections, especially its methicillin-resistant strains (MRSA)[23]. Using AlphaFold3[24], we predicted the structure of TGTs from *S. aureus* and *E. coli*. Despite only 35% sequence identity to EcTGT (Figure S1), the predicted structures of SaTGT and EcTGT were largely superimposable, with an RMSD of 0.720 Å. However, SaTGT features an expanded cavity in its active site (Figure S2), with a predicted volume of 348.62 Å³ for EcTGT and 778.25 Å³ for SaTGT. The increased size of the active site cavity suggested the possibility that SaTGT might exhibit greater tolerance for nonideal substrates such as DNA. It should be noted that limited work has been performed in characterizing the structure and activity of TGTs from different organisms. Most studies describing the enzymatic properties of TGT have used the EcTGT as a model, and to date, the structure of prokaryotic TGTs have been described only for *Zymomonas mobilis* and *Thermotoga maritima*[25,26].

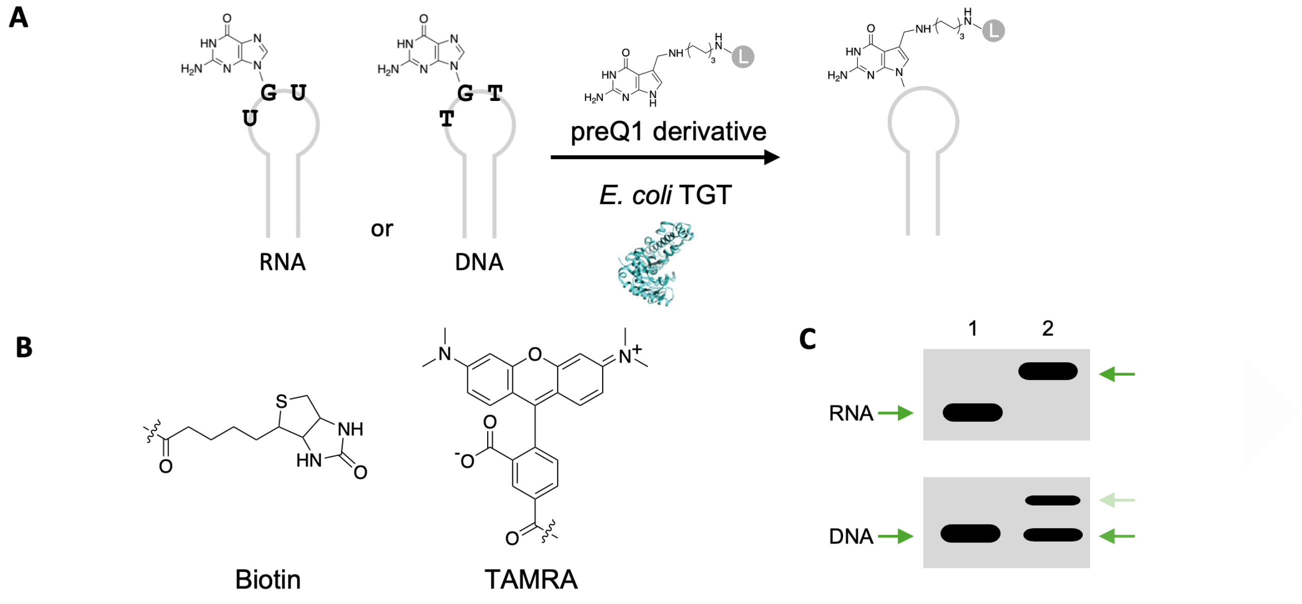

Figure 1. EcTGT mediated RNA or DNA modification with functionalized small molecules. (A) General scheme of RNA-TAG and DNA-TAG. EcTGT irreversibly exchanges a specific guanine with a preQ1-ligand (L) on an RNA or DNA hairpin. (B) preQ1-ligands used in this work include preQ1-biotin, preQ1-tetramethylrhodamine (TAMRA). (C) Cartoon depiction of expected upward gel shift due to increased mass from the base exchange reaction as indicated by the green arrows. DNA labeling by EcTGT is less efficient than RNA labeling under the same condition. Lane 1: DNA or RNA start material; Lane 2: preQ1-L modified DNA or RNA.

SaTGT was recombinantly expressed and purified in *E. coli*. To our knowledge, this is the first reported production of this enzyme. (Figure S3). We first sought to test whether purified SaTGT exhibited activity towards natural tRNA substates from *S. aureus in vitro*. In an RNA labeling assay using TAMRA-conjugated preQ1, our results showed that the SaTGT effectively labeled both tRNA^{Tyr} and tRNA^{His} with the TAMRA probe, confirming isolation of active enzyme (Figure 2A). We next tested SaTGT against a minimal recognition element based on *E. coli* tRNA^{Tyr}, the 25-mer RNA hairpin ECYMH commonly used in our past RNA-TAG studies. The result showed that SaTGT successfully incorporated preQ1-biotin onto ECYMH, although with slightly lower efficiency than EcTGT. Previous studies showed that EcTGT is unable to modify the DNA equivalent substrate (dECYMH), and we wondered if SaTGT showed more tolerance for the dECYMH substrate. Interestingly, SaTGT exhibited markedly superior labeling of dECYMH, whereas EcTGT showed no appreciable modification (Figure 2B). From an evolutionary perspective, SaTGT may have greater compatibility with RNA or DNA hairpins derived from its native tRNAs. We therefore reasoned that DNA hairpin substrates derived from these alternative *S. aureus* tRNAs would be preferable substrates for labeling reactions. Following established nomenclature, we designated the 25-nuceotide wide-type hairpins as dSAH-0, dSAY-0, dSAD-0, dSAN-0, corresponding to the hairpins derived from tRNA^{His}, tRNA^{Tyr}, tRNA^{Asp}, and tRNA^{Asn} from *S. aureus*, respectively. Although all four hairpins share an identical UGU recognition sequence, the flanking anti-codon loop residues differ significantly. Each hairpin was tested in a labeling reaction using a preQ1-biotin label and both EcTGT and SaTGT. Interestingly, both TGTs showed clear labeling of dSAH-0 but virtually no activity toward the other three substrates. SaTGT showed significantly higher labeling efficiency towards DNA than EcTGT (Figure 2C). This preference for tRNA^{His} derived DNA hairpins mirrors our previous findings with EcTGT and suggested that this hairpin was the most promising scaffold for further optimization[12].

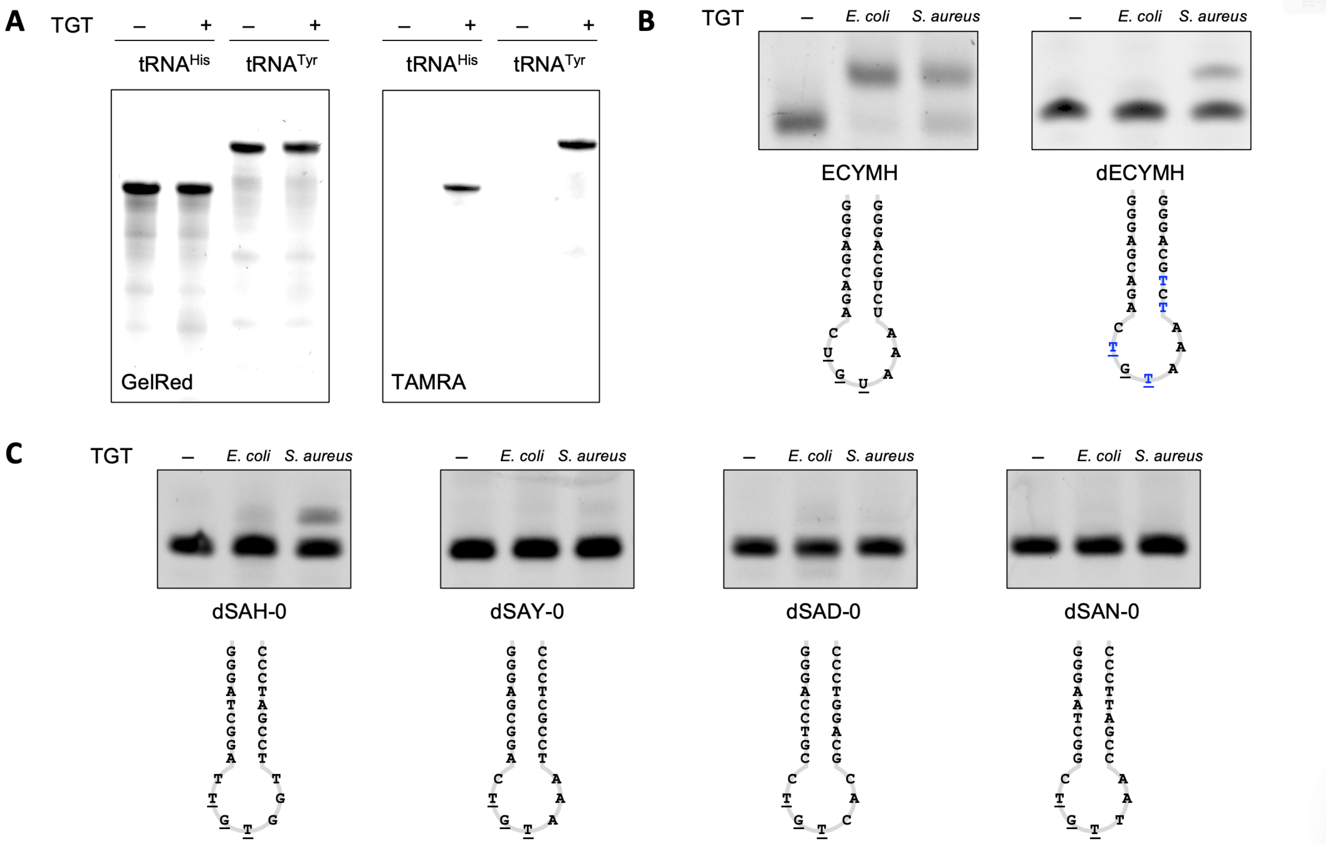

Figure 2. TGT-mediated labeling of tRNAs and hairpin substrates. (A) SaTGT can label its natural tRNA substrates with modified ligand preQ1-TAMRA. (B) EcTGT and SaTGT labeling of RNA hairpin ECYMH and its DNA counterpart dECYMH using preQ1-biotin. While both SaTGT and EcTGT can modify the ECYMH RNA hairpin, only SaTGT showed labeling for the corresponding DNA-hairpin (dECYMH). (C) EcTGT and SaTGT labeling of small DNA hairpins derived from cognate tRNA substrates from *S. aureus* with preQ1-biotin. SaTGT showed clear labeling of $tRNA^{His}$-derived hairpin dSAH-0.

Encouraged by our preliminary findings, we decided to use dSAH-0 as a starting point for further optimization to achieve higher DNA labeling efficiency. Previous studies of EcTGT demonstrated that low DNA labeling efficiency is likely a result of steric hindrance, and that the anti-codon loop sequence is the dominant determinant of recognition and efficiency[12]. We designed new DNA hairpin substrates by introducing loop mutations replacing larger purine bases with smaller pyrimidines (C or T) and exploring random combinations of these bases to reduce steric clashes, while keeping the stem region sequence identical to the *S. aureus* $tRNA^{His}$. Among the variants tested, dSAH-6 exhibited the highest labeling efficiency (>80%) with preQ1-biotin and was selected for all subsequent experiments (Figure 3A, B). This result further supports our previous conclusion that steric effects play a major role in substrate recognition, extending its relevance beyond EcTGT to SaTGT and potentially other orthologs[12].

For practical DNA-TAG applications, the TGT recognition hairpin must be compatible with diverse positions within longer DNA constructs. We therefore designed 67-mer single-stranded DNA oligonucleotides containing the dSAH-6 sequence at the 5′ end, 3′ end, or at an internal site. All three constructs were quantitatively labeled with preQ1-biotin by SaTGT (Figure 3C), confirming the versatility of the optimized hairpin. To enable applications requiring enhanced sensitivity, we explored whether SaTGT could label multiple sites within the same oligonucleotide. Constructs bearing two tandem dSAH-6 hairpins were fully labeled by SaTGT, indicated by observation of a single band of labelled product with a higher gel shift compared to a control containing a single active hairpin and one inactivated by a TGT→TCT mutation which showed a lesser gel shift (Figure 3D). In contrast, EcTGT typically yields incomplete labeling when the recognition hairpin is placed at the 5′ terminus, highlighting the superior positional tolerance of DNA labelling using SaTGT[12].

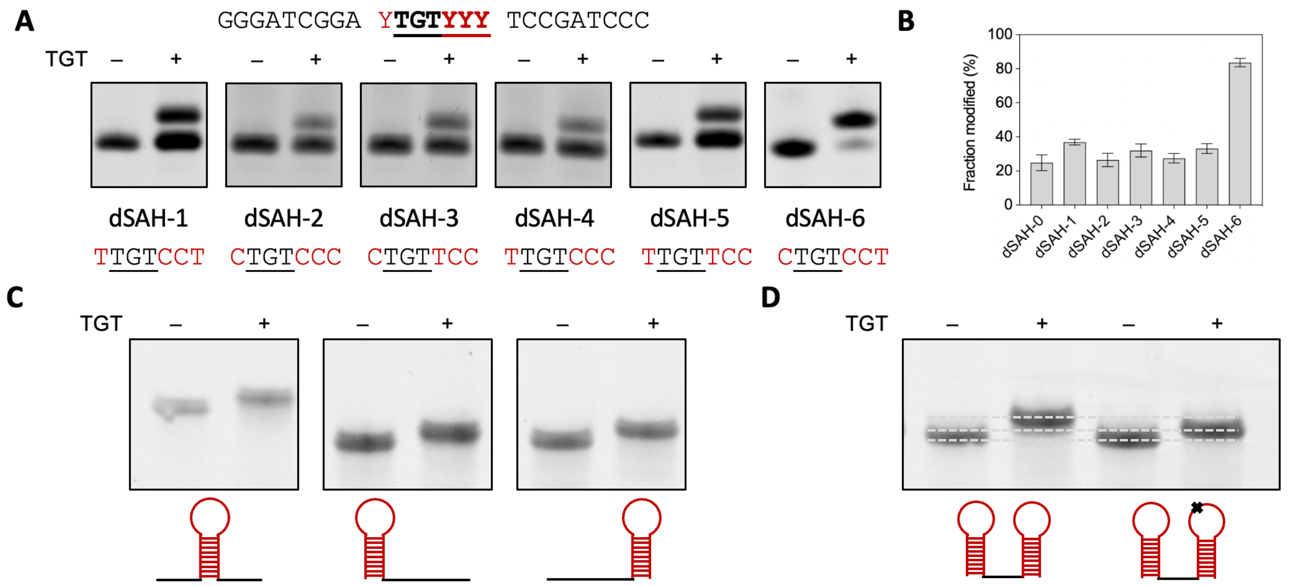

Figure 3. SaTGT mediated labeling of DNA. (A) SaTGT labeling of mutants. Unmodified oligos are on the left, and reaction products are on the right. Modification is determined by an upward gel shift of the oligo after insertion of preQ1-biotin. (B) Quantification of labeling efficiency for mutants. SaTGT modifies dECY-6 at ~80% efficiency. (C) dSAH-6 construct placement. Efficient labeling occurs at either end, or internally, within the DNA construct. (D) Insertion of multiple labels within a single construct. Multiple probes can be incorporated in tandem for enhanced sensitivity, confirmed by increased gel shift as compared to ΔC mutation indicated as X. The bottom dashed line represents unlabeled ssDNA, the middle line single-labeled ssDNA, and the top line double-labeled DNA with larger band upshift.

To directly compare the performance of SaTGT and EcTGT, we conducted parallel labeling experiments. Against 25-mer DNA substrates, SaTGT exhibited superior labeling efficiency not only with its own optimized substrate dSAH-6 but also with dECH-10, a DNA hairpin sequence previously optimized for EcTGT (Figure 4A, B). We note that the 25-mer extended dECH-10 hairpin (TAG3) we used, exhibited lower efficiency of labeling compared to the shorter 17-mer dECH-10 hairpin (TAG2)[12]. To gain deeper insight into their kinetic differences, we performed time-course experiments with two commonly used preQ1 derivatives. For both preQ1-biotin and preQ1-TAMRA, SaTGT displayed significantly faster labeling kinetics and higher final yields than EcTGT on the respective optimized DNA hairpins. For preQ1-biotin, the half-labeling time ($t_{1/2}$) for reaching saturation was around 45 min for SaTGT, whereas EcTGT showed only relative weak labeling under the same experimental conditions (Figure 4C). For preQ1-TAMRA, SaTGT showed around 3.5-fold faster initial rate according to the TAMRA fluorescence signals (Figure S4) and 1.5-fold higher final labeling efficiency than EcTGT (Figure 4D). These results establish SaTGT as a more effective enzyme for DNA-TAG applications.

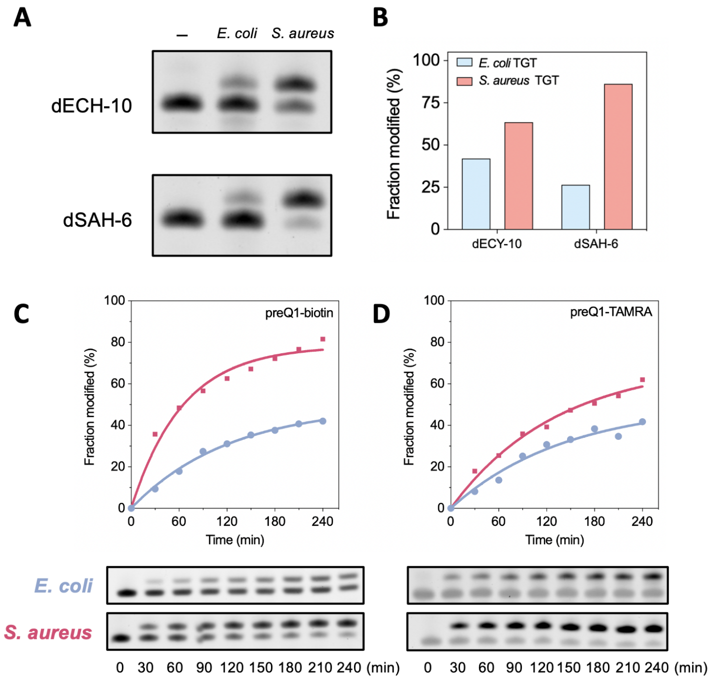

Figure 4. Comparison of EcTGT and SaTGT for DNA labeling. (A) Labeling of dECH-10 and dSAH-6 hairpins with preQ1-biotin by EcTGT and SaTGT. SaTGT exhibits higher efficiency against both substrates. (B) Quantification of labeling efficiency from (A). (C) Labeling kinetics of preQ1-biotin. (D) Labeling kinetics of preQ1-TAMRA. SaTGT exhibits significantly faster kinetics and higher final yields than EcTGT on both optimized DNA substrates.

Conclusion

In this study, we report the first expression and purification of active TGT from *S. aureus and* demonstrate its superior performance for site-specific covalent DNA labeling. We discovered that SaTGT exhibits markedly higher tolerance for DNA hairpin substrates than the more widely used EcTGT. These findings illustrate how natural variation in TGT substrate promiscuity can be directly harnessed to overcome limitations in substrate labeling. SaTGT significantly expands the DNA-TAG toolbox, transforming a previously inefficient reaction into a practical and versatile technology. The new enzyme opens exciting opportunities for nucleic acid diagnostics, therapeutics, imaging, nanotechnology, and chemical barcoding applications[27–29]. Further screening of TGTs from diverse species may yield additional variants with unique properties, enriching enzymatic nucleic acid modification strategies.

Supporting Information

The authors have cited additional references within the Supporting Information[22–24].

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Keywords: DNA labeling • DNA • tRNA Guanine Transglycosylase • *S. aureus* • *E. coli*

[1] N. Klöcker, F. P. Weissenboeck, A. Rentmeister, "Covalent labeling of nucleic acids" *Chem Soc Rev* **2020**, *49*, 8749–8773.

[2] T.-K. Kha, Y. Zhao, R.-Y. Zhu, "Site-Selective Modification and Labeling of Native RNA" *Chemistry – A European Journal* **2025**, *31*, e202404244.

[3] D. Gillingham, R. Shahid, "Catalysts for RNA and DNA modification" *Curr Opin Chem Biol* **2015**, *25*, 110–114.

[4] J. M. Holstein, A. Rentmeister, "Current covalent modification methods for detecting RNA in fixed and living cells" *Methods* **2016**, *98*, 18–25.

[5] P. Le, N. Ahmed, G. W. Yeo, "Illuminating RNA biology through imaging" *Nat Cell Biol* **2022**, *24*, 815–824.

[6] L. Li, S. Xu, H. Yan, X. Li, H. S. Yazd, X. Li, T. Huang, C. Cui, J. Jiang, W. Tan, "Nucleic Acid Aptamers for Molecular Diagnostics and Therapeutics: Advances and Perspectives" *Angew Chem Int Ed Engl* **2021**, *60*, 2221–2231.

[7] B. Stengl, K. Reuter, G. Klebe, "Mechanism and Substrate Specificity of tRNA–Guanine Transglycosylases (TGTs): tRNA-Modifying Enzymes from the Three Different Kingdoms of Life Share a Common Catalytic Mechanism" *ChemBioChem* **2005**, *6*, 1926–1939.

[8] J. Díaz-Rullo, J. E. González-Pastor, "tRNA queuosine modification is involved in biofilm formation and virulence in bacteria" *Nucleic Acids Research* **2023**, *51*, 9821–9837.

[9] S. Nakanishi, T. Ueda, H. Hori, N. Yamazaki, N. Okada, K. Watanabe, "A UGU sequence in the anticodon loop is a minimum requirement for recognition by Escherichia coli tRNA-guanine transglycosylase" *J Biol Chem* **1994**, *269*, 32221–32225.

[10] J. K. Hurt, S. Olgen, G. A. Garcia, "Site-specific modification of Shigella flexneri virF mRNA by tRNA-guanine transglycosylase in vitro" *Nucleic Acids Res* **2007**, *35*, 4905–4913.

[11] A. W. Curnow, G. A. Garcia, "tRNA-guanine transglycosylase from *Escherichia coli*: Recognition of dimeric, unmodified tRNATyr" *Biochimie* **1994**, *76*, 1183–1191.

[12] E. M. Tota, N. K. Devaraj, "Site-Specific Covalent Labeling of DNA Substrates by an RNA Transglycosylase" *J. Am. Chem. Soc.* **2023**, *145*, 8099–8106.

[13] D. Zhang, L. Liu, S. Jin, E. Tota, Z. Li, X. Piao, X. Zhang, X.-D. Fu, N. K. Devaraj, "Site-Specific and Enzymatic Cross-Linking of sgRNA Enables Wavelength-Selectable Photoactivated Control of CRISPR Gene Editing" *J. Am. Chem. Soc.* **2022**, *144*, 4487–4495.

[14] D. Zhang, C. Y. Zhou, K. N. Busby, S. C. Alexander, N. K. Devaraj, "Light-Activated Control of Translation by Enzymatic Covalent mRNA Labeling" *Angew Chem Int Ed* **2018**, *57*, 2822–2826.

[15] K. N. Busby, A. Fulzele, D. Zhang, E. J. Bennett, N. K. Devaraj, "Enzymatic RNA Biotinylation for Affinity Purification and Identification of RNA–Protein Interactions" *ACS Chem. Biol.* **2020**, *15*, 2247–2258.

[16] E. M. Tota, N. K. Devaraj, "RNA-TAG Mediated Protein-RNA Conjugation" *ChemBioChem* **2023**, *24*, e202300454.

[17] C. Y. Zhou, S. C. Alexander, N. K. Devaraj, "Fluorescent turn-on probes for wash-free mRNA imaging via covalent site-specific enzymatic labeling" *Chem. Sci.* **2017**, *8*, 7169–7173.

[18] H. Chen, D. Liu, A. Aditham, J. Guo, J. Huang, F. Kostas, K. Maher, M. J. Friedrich, R. J. Xavier, F. Zhang, X. Wang, "Chemical and topological design of multicapped mRNA and capped circular RNA to augment translation" *Nat Biotechnol* **2025**, *43*, 1128–1143.

[19] E. M. Tota, Enzymatic Tools for Enabling Nucleic Acid Driven Technologies, UC San Diego, **2021**.

[20] Y. Yuan, R. Zallot, T. L. Grove, D. J. Payan, I. Martin-Verstraete, S. Šepić, S. Balamkundu, R. Neelakandan, V. K. Gadi, C.-F. Liu, M. A. Swairjo, P. C. Dedon, S. C. Almo, J. A. Gerlt, V. de Crécy-Lagard, "Discovery of novel bacterial queuine salvage enzymes and pathways in human pathogens" *Proceedings of the National Academy of Sciences* **2019**, *116*, 19126–19135.

[21] S. Quaiyum, Y. Yuan, G. Sun, R. M. M. N. Ratnayake, G. Hutinet, P. C. Dedon, M. F. Minnick, V. de Crécy-Lagard, "Queuosine salvage in Bartonella henselae Houston 1: a unique evolutionary path" *Microbiology* **2024**, *170*, 001490.

[22] S. C. Alexander, K. N. Busby, C. M. Cole, C. Y. Zhou, N. K. Devaraj, "Site-Specific Covalent Labeling of RNA by Enzymatic Transglycosylation" *J. Am. Chem. Soc.* **2015**, *137*, 12756–12759.

[23] R. Touaitia, A. Mairi, N. A. Ibrahim, N. S. Basher, T. Idres, A. Touati, "Staphylococcus aureus: A Review of the Pathogenesis and Virulence Mechanisms" *Antibiotics* **2025**, *14*, 470.

[24] J. Abramson, J. Adler, J. Dunger, R. Evans, T. Green, A. Pritzel, O. Ronneberger, L. Willmore, A. J. Ballard, J. Bambrick, S. W. Bodenstein, D. A. Evans, C.-C. Hung, M. O'Neill, D. Reiman, K. Tunyasuvunakool, Z. Wu, A. Žemgulytė, E. Arvaniti, C. Beattie, O. Bertolli, A. Bridgland, A. Cherepanov, M. Congreve, A. I. Cowen-Rivers, A. Cowie, M. Figurnov, F. B. Fuchs, H. Gladman, R. Jain, Y. A. Khan, C. M. R. Low, K. Perlin, A. Potapenko, P. Savy, S. Singh, A. Stecula, A. Thillaisundaram, C. Tong, S. Yakneen, E. D. Zhong, M. Zielinski, A. Žídek, V. Bapst, P. Kohli, M. Jaderberg, D. Hassabis, J. M. Jumper, "Accurate structure prediction of biomolecular interactions with AlphaFold 3" *Nature* **2024**, *630*, 493–500.

[25] R. Brenk, M. T. Stubbs, A. Heine, K. Reuter, G. Klebe, "Flexible adaptations in the structure of the tRNA-modifying enzyme tRNA-guanine transglycosylase and their implications for substrate selectivity, reaction mechanism and structure-based drug design" *Chembiochem* **2003**, *4*, 1066–1077.

[26] K. Reuter, R. Ficner, "RNA-modification by Base Exchange: Structure, Function and Application of tRNA-guanine Transglycosylases" *Journal of Molecular Biology* **2025**, *437*, 168980.

[27] D. T. Flood, S. Asai, X. Zhang, J. Wang, L. Yoon, Z. C. Adams, B. C. Dillingham, B. B. Sanchez, J. C. Vantourout, M. E. Flanagan, D. W. Piotrowski, P. Richardson, S. A. Green, R. A. Shenvi, J. S. Chen, P. S. Baran, P. E. Dawson, "Expanding Reactivity in DNA-Encoded Library Synthesis via Reversible Binding of DNA to an Inert Quaternary Ammonium Support" *J. Am. Chem. Soc.* **2019**, *141*, 9998–10006.

[28] D. T. Flood, C. Kingston, J. C. Vantourout, P. E. Dawson, P. S. Baran, "DNA Encoded Libraries: A Visitor's Guide" *Israel Journal of Chemistry* **2020**, *60*, 268–280.

[29] F. Hövelmann, I. Gaspar, S. Loibl, E. A. Ermilov, B. Röder, J. Wengel, A. Ephrussi, O. Seitz, "Brightness through Local Constraint—LNA-Enhanced FIT Hybridization Probes for In Vivo Ribonucleotide Particle Tracking" *Angewandte Chemie International Edition* **2014**, *53*, 11370–11375.

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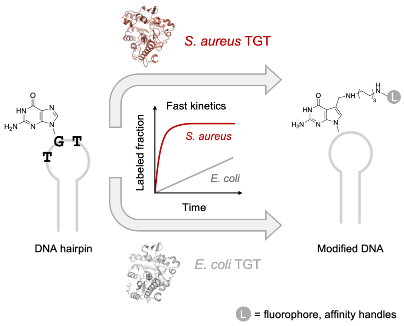

Here we present *Staphylococcus aureus* tRNA guanine transglycosylase (SaTGT) as an effective enzyme for site-specific covalent DNA labeling. By rationally optimizing a $tRNA^{His}$-derived DNA hairpin, SaTGT achieves high-yield incorporation of preQ1 derivatives. We demonstrate the optimized DNA substrate enables rapid single- and multi-site labeling at multiple positions in ssDNA constructs, expanding the DNA-TAG (transglycosylation at guanosine) toolbox for imaging, diagnostics and therapeutics.